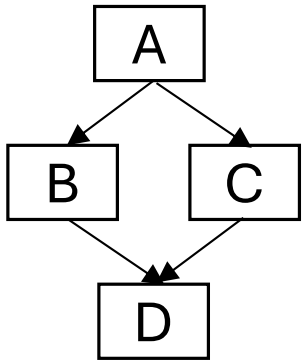
More fork-join

5/12/25

Recall: fork and join

```
A();
fork B();
C();
join;
D();
```



Each time unit, each processor runs a strand w/o unfinished predecessors (if one exists) and finishes it

Both B and C must run after A and before D, but their order is flexible

Which of the following could be printed by the code below?

```
System.out.print(1);
fork System.out.print(2);
System.out.print(3);
System.out.print(4);
join;
```

I. 1234
II. 2134
III. 1324

A. Only I.
B. Only II.
C. Only I. and II.
D. Only II. and III.
E. None of the above

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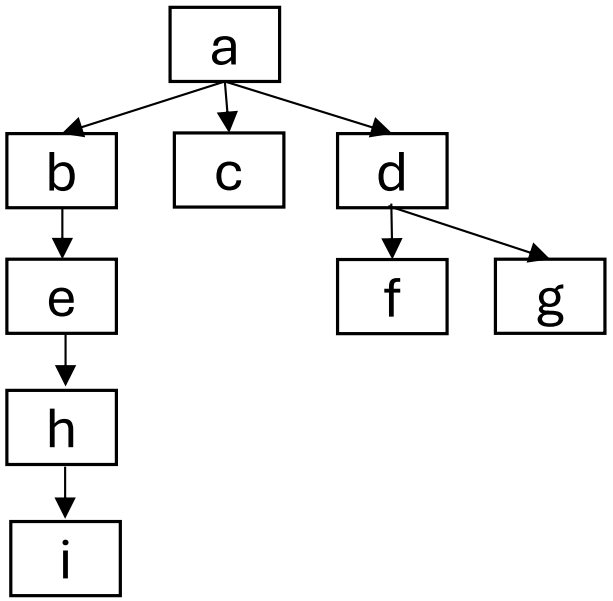
How long will this graph run on 2 processors?



A. Exactly 4 time units
B. Exactly 5 time units
C. Either 4 or 5 time units
D. Either 5 or 6 time units
E. None of the above

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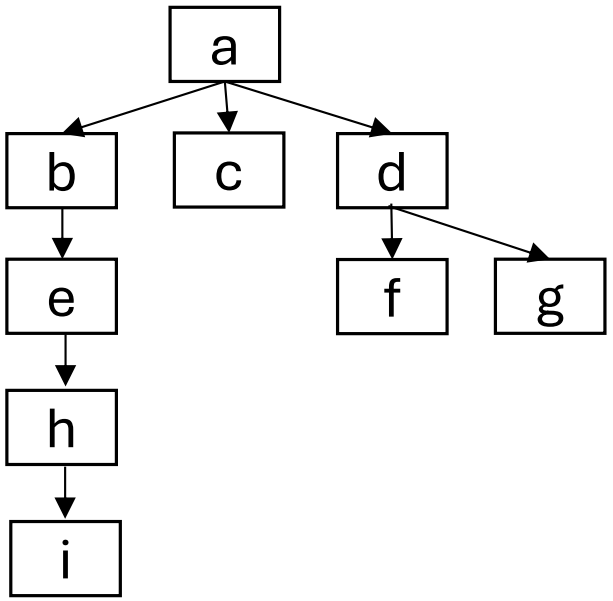

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(between 5 and 7)

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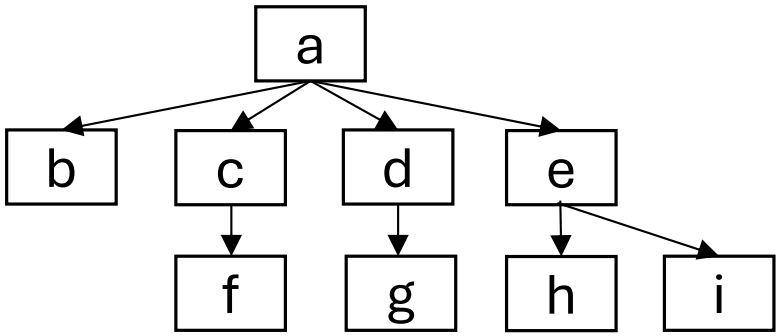

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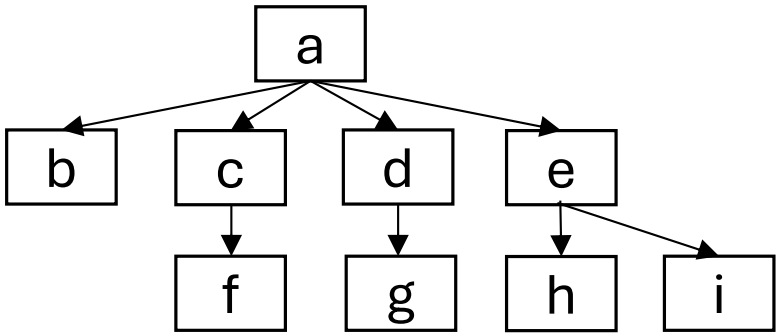

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What are the work and span of this graph?

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C. 6 and 3
D. 7 and 3
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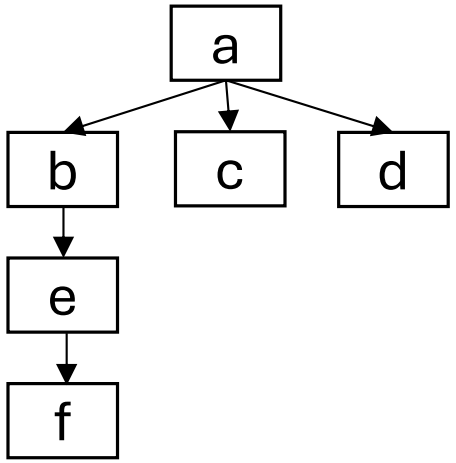

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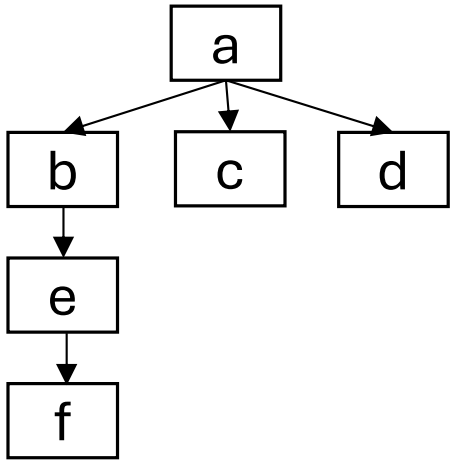
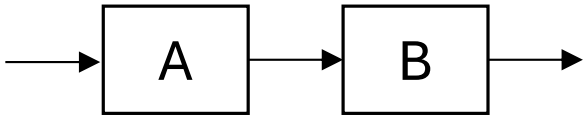
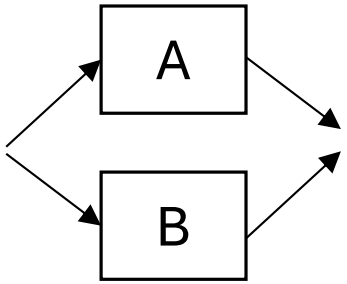


Combining subcomputations





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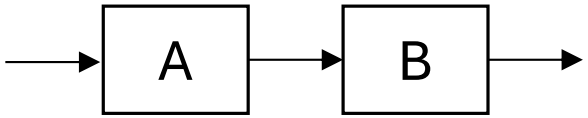

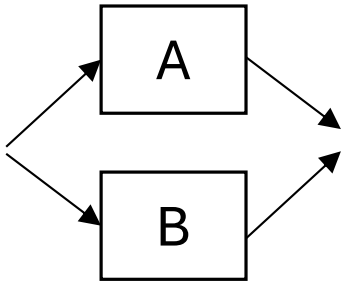

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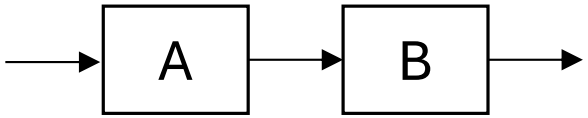

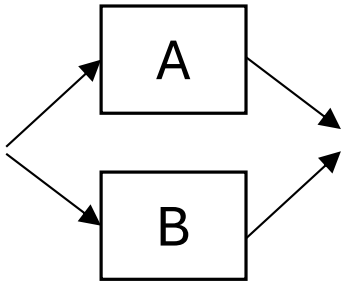

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Span: $T_\infty(A) + T_\infty(B)$

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Span: $\max\{ T_\infty(A), T_\infty(B) \}$

What best characterizes the span of this code?

(n is hi-low)

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int sum(int[] array, int low, int hi) {
        if(hi==low+1)
                return array[low];
        int left = fork sum(array, low, (hi+low)/2);
        int right = sum(array, (hi+low)/2, hi);
        join;
        return left + right;
}
```

A. O(1)
B. O(log n)
C. O(n)
D. O(n log n)
E. None of the above

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What best characterizes the work and span of this code?

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```
int sum(int[] array, int low, int hi) {
        int total = 0;
        for(int i=low; i < hi; i++)
                total += array[i];
        return total;
}
```

A. O(log n) and O(log n)
B. O(log n) and O(n)
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In actual Java: RecursiveTask

```
class Sum extends RecursiveTask<Integer> {
    //attributes and constructor

    protected Integer compute(){
        if(hi – low < THRESHOLD) return simpleSum(array, low, hi);
        Sum left = new Sum(array, low, (hi+low)/2);
        Sum right= new Sum(array, (hi+low)/2, hi);
        left.fork();
        return right.compute() + left.join();
    }
}
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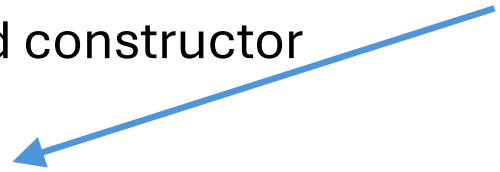

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Only recurse to create enough strands

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    }
}
```

Note: Only create new strand for one half

In actual Java: main

```
sum = ForkJoinPool.commonPool().invoke(
                        new Sum(array, 0, array.length));
```